Lab 15: Macroevolution (Module 14)

BIOL-1

Name: _____ **Date:** _____

Learning Objectives

By the end of this lab, you should be able to:

1. Apply the biological species concept to **predict** when **gene flow** can stop or reverse lineage splitting.
2. Classify **prezygotic** vs **postzygotic** barriers and name sub-types from **messy** narratives.
3. Contrast **allopatric** vs **sympatric** speciation with **plausible** mechanisms.
4. Explain how microevolutionary change can add up to **reproductive isolation**.

Overview

Where this sits: In **Module 13 / Lab 14**, you followed **allele frequencies** in **one** population (microevolution). In **Module 14 / Lab 15**, the question shifts: **When does one lineage become two?** Reproductive **isolation** and **reduced gene flow** are the stars.

You are on the **Speciation Response Team**. A river canyon has split one ancestral **mock-orange fly** (fictional species for this worksheet) into **Population North** and **Population South**. Your job is to **predict** what must happen for **one species to become two**, then diagnose **barriers** from field reports, and finally decide whether a weird plant counts as **quick speciation**. This lab is a chain of **if–then** puzzles on paper—not a vocabulary list to memorize out of order.

Terms you will use (one glance)

Term	In one line
Biological species concept (BSC)	Groups are often treated as one species if they **interbreed** (or could) and produce viable/fertile offspring; **ongoing gene flow** tends to keep them one line.
Reproductive isolation	**Barriers** that make **interbreeding** rare or **hybrids** unsuccessful—**prezygotic** (before a zygote) or **postzygotic** (after).
Prezygotic	Mating, timing, or gametes "don't meet" the way you'd need for a zygote (first line of **no gene flow**).
Postzygotic	A zygote or hybrid has trouble surviving or **reproducing** (e.g. embryo dies, sterile hybrid).
Allopatric	Isolation that starts with a **geographic** split.
Sympatric	Speciation in the **same** area (e.g. **instant** isolation from a genome change in one plant).

Materials Needed

- Pencil or pen
- This worksheet

Part 1: Canyon Split — Build a Speciation Hypothesis

One species, two valleys, no bridges. Flies rarely cross the canyon.

Road map: A. Predict (before the "report") → **B. Data** (read the field report) → **C. Revise** (fix your first idea with evidence).

> **A. Prediction:** If North and South stay isolated for thousands of generations, **mutations and selection** can proceed independently. **Before you read "Report B,"** answer:

Hypothesis: Will the two populations *necessarily* become two separate species **automatically** just because they end up **looking** different? **Yes / No.**
Hint: The biological species concept is about mating, offspring, and gene flow—not about looks alone. Looking different = morphology; it does not, by itself, "clock in" as two BSC species.

Reason:

B. Report (given): After 10,000 generations, North flies breed in **early spring**; South flies breed in **late spring**. When researchers **cage** them together, mating is **rare**; when they **force** mating, eggs are fertilized but **embryos die** before hatching.

1. List two distinct reproductive barriers in that report (name the **types**—e.g. temporal, reduced hybrid viability, etc.):

1.
2.

2. Which barrier is prezygotic? Which is postzygotic?

- Prezygotic:
- Postzygotic:

C. Revision: Did your Part A idea need to change? What **evidence** made you tighten or fix it?

Part 2: Mystery Reports — Diagnose the Barrier

Each story is incomplete on purpose—your call on Pre vs Post and sub-type is the point.

3. For each report, mark **Pre** (prezygotic) or **Post** (postzygotic).

Part 2 — Prezygotic or postzygotic

Field report	Pre or Post?
Pollen of Species X never starts tubes on Species Y stigmas.	
F1 hybrid salamanders survive and mate, but produce **no functional gametes**.	
Fireflies flash different patterns; males and females ignore the wrong pattern.	
Hybrid zygotes form, but chromosome pairing fails in early mitosis so development stops.	

4. Sub-type: Match each line of **question 3** to the best label. Choose from: *habitat* / *temporal* / *behavioral* / *mechanical* / *gametic* / *reduced hybrid viability* / *reduced hybrid fertility* / *hybrid breakdown*.

Part 2 — Barrier sub-type

Report (same order as in Q3)	Barrier sub-type
Pollen of Species X never starts tubes on Species Y stigmas.	
F1 hybrid salamanders survive and mate, but produce **no functional gametes**.	
Fireflies flash different patterns; males and females ignore the wrong pattern.	
Hybrid zygotes form, but chromosome pairing fails in early mitosis so development stops.	

Stretch (Part 2). Short answers—use the same four reports if helpful.

(a) In one sentence: why does **behavioral isolation** (e.g. different firefly flash patterns) block gene flow **before** a zygote counts as "formed" for the pre/post split?

(b) The **F1 salamanders** that survive and mate but produce no functional gametes: state **prezygotic or postzygotic** and the **sub-type** label you would use.

(c) Suppose hybrids between two plant species **germinate** but **die** as seedlings before they flower. Is the main barrier **pre-** or **postzygotic**, and why?

Part 3: Geography — Same River, Different Endings

Polyploidy in one line: A cell can get **extra full sets of chromosomes** (e.g. **4n** has twice the "library" of a typical **2n**). A new **4n** plant can be reproductively isolated from **2n** relatives in the same meadow. For a lasting lineage to form, it must also reproduce, for example through compatible **4n** mates or self-fertilization. Chromosome doubling alone does not guarantee an established species.

5. Two roads to new species:

Part 3 — Allopatric or sympatric

Scenario	Allopatric or Sympatric?
Canyon splits a valley; populations diverge in **isolation**.	
A plant instantly inherits **4n** chromosomes and can only breed with other **4n** plants in the **same** meadow.	

6. Mechanism check: Why is polyploidy often **sympatric** speciation in plants? *(Hint: "instant" isolation from the diploid population in the same field.)*

7. Counterfactual: If the canyon **fills in** and the flies **mix** freely **before** strong barriers evolve, what happens to **this** speciation story? **One paragraph (4–5 sentences):**

Part 4: From Lab 14 to Today — Micro Builds Macro

No new "evolution force" appears at the speciation line—barriers to gene flow are what you add to the microevolution you already know.

8. Chain argument: Starting from **mutation** creating variation in each isolated population, explain how **natural selection** and **drift** could make **prezygotic** barriers more likely over time. (4–5 sentences)

9. Final verdict: Two groups occasionally produce **fertile hybrids** in a laboratory. Is that enough to call them one species or two under the biological species concept? Explain what evidence about **mating and gene flow in nature** you still need. Do not treat a single hybrid as a complete species test.

10. Pace of splitting: Which path in this lab—**many generations** in separated valleys (**allopatric** mock-orange flies) or **one-generation** polyploidy in the **same** meadow—typically needs **more time** for **independent** evolution before the groups are reproductively isolated? **One short paragraph.**

11. BSC limits: Name **one situation** where the **biological species concept** is **hard to apply** in the field (hint: fossils-only lineages, bacteria, or strictly asexual groups). **Two or three sentences.**

Wrap-up

W1. Name **one** barrier you almost mixed up (pre vs post, or two sub-types) and what finally **clicked**.

W2. Optional fun: One **real** or **movie/TV** example of **isolation** (courtship, timing, or hybrid trouble)—**one** sentence, no need for perfect Latin names.

W3. Looks aren't enough: When would using **appearance alone** be a **misleading** way to decide if two populations are **one** species or **two**?

W4. Micro → macro: In **Lab 14** you tracked **allele frequencies** in one population. If **migration** between North and South mock-orange flies were **zero** for a long time, how does that help the **two valleys** accumulate **different** genetic changes? **2–3 sentences.**

Lab 15 — Macroevolution. Speciation Response Team scenario (paper lab).